Designing a Solar/Wind Hybrid Power System for Charging Electric Vehicles

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***Abstract*—Electric vehicles (EVs) are becoming more prevalent in modern society. The residential power outlet may be used to charge the EVs using the AC charger. However, on interstate highways, a power plant using renewable sources can be used for automobiles when long driving is required. Renewable energy sources like solar PV systems and wind turbines are strongly advised for use in EV applications. These non-polluting sources generate power, which is used for propulsion and stored in batteries. The idea of charging EVs utilizing a local hybrid solar/wind power system in Lubbock has been presented in this paper. The PV and wind farms are linked to EV stations using power converters. Furthermore, in the time of peak demand, the proposed system can be connected to the grid to balance the load demand. Simulation results show the feasibility and the potential of the proposed system.**

Keywords—Wind turbine, Solar photovoltaic system, Electric vehicles

I. Introduction

Using an electric vehicle can minimize the damages that the transportation section has on the environment. The cost, charging structure, power electrical system, and battery capacity are all factors that affect how well EVs work [1]. The utilization of ecologically friendly energy sources like solar and wind energy is inevitable in light of global warming and the potential for fossil fuel shortages. Utilizing solar and wind energy has the benefit of being cheap, plentiful, pollution-free, and available everywhere. The battery has to be charged in order for the car to operate. No need to mention that the car requires more charge while traveling over long distances. As a result, charging facilities for EVs are situated along the sides of the highway. These automobiles are charged with the help of power produced by renewable energy sources, which are close to the charging station [2]. Furthermore, the energy from various input sources had to be transferred via power converters, including isolated and non-isolated types. Isolated converters are more sophisticated than non-isolated converters due to the use of transformers. As a result, transformers are not required in the circuits of the majority of applications, which instead require a multiport dc-dc converter [3]. In addition, in order to extract maximum power from the solar part of the system, there is a need for a tracking technique. In this paper, the P&O MPPT technique is used to extract the most power possible from the PV component [4]. The main contribution of this paper is to charge EVs with renewable energies both in the daytime and nighttime. This is how the paper will be presented: The characteristics of the solar/wind system are presented in the next part. The suggested topology is described in part 3. The simulation results are shown in part 4. And finally, the conclusion is in the last part.

II. Solar/wind system characteristics

In order to explain and build the hybrid system presented in this paper, solar PV panels and wind turbines combine to create a highly reliable hybrid energy system that can harvest renewable energy under various weather situations [5, 6]. When the sun is shining and the wind isn't too strong, solar panels will generate energy; nevertheless, when severe weather strikes and the sky is cloudy, wind turbines become crucial. To maximize the utilization of renewable energy resources, the suggested solar/wind hybrid power systems combine both sources of electricity with a bank of batteries to store the energy. In the case of extra energy, by using an inverter, the extra energy from the solar part can feed the power grid [7, 8]. In addition, to design a power plant, a machine learning model is required to predict future energy [9-11].

1) Solar PV subsystem

Solar PV systems use the photovoltaic effect to turn sun radiation into power. Semiconductor materials with P-N junctions employ this method to generate energy when exposed to light [12]. Fig 1 shows a solar cell's equivalent circuit.

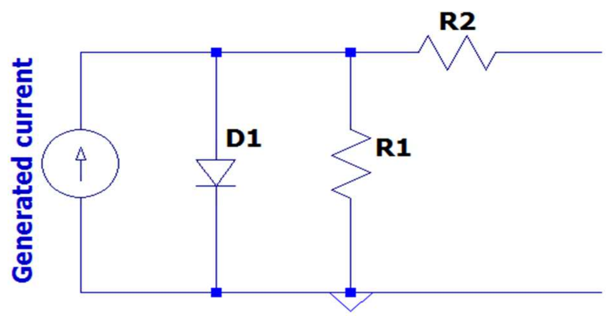

Fig. 1. solar cell's equivalent circuit

The generated current in the secondary side of each solar panel is obtained as follow:

$$I_o = I_{sh} - I_s \left(\exp\left(\frac{V+IR_2}{nV_t}\right) - 1 \right) - \frac{V_o+IR_2}{R_1} \quad (1)$$

Where Vo and Io are the generated voltage and current. I-V curves for solar cells show how current changes with voltage in a photovoltaic device. Characteristics of Solar Cells I-V Curves are essentially a visual summary of a module's operation that shows how current, and voltage relate to each other under the different irradiance and temperature circumstances [13]. Fig. 2 depicts the I-V, P-V curves of the proposed solar module.

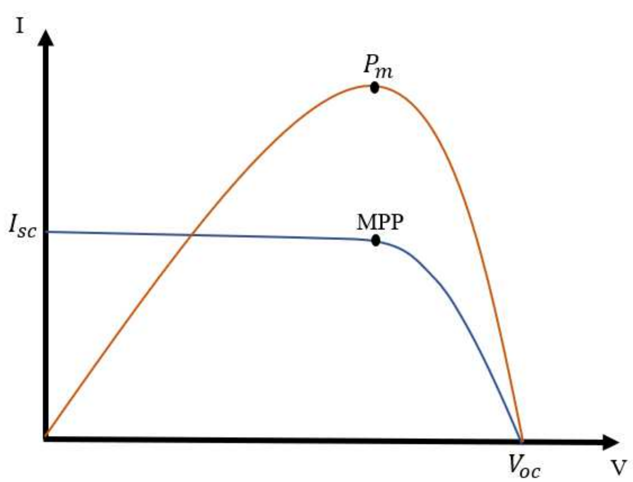

Fig. 2. Solar module I-V and P-V characteristic

An MPPT approach with a buck-boost converter is required in order to obtain the highest amount of energy from the sun. The P&O approach is used in this paper to track the maximum power points at a different time of the day. If the output power of the system is more than the needed energy for the charging station, as a demand side management with utilizing a multilevel inverter, the extra power can feed the power grid [14, 15].

2) Wind turbines subsystem

Recent years have seen the emergence of wind energy due to the quick development of new technology and wind turbines. A rotor, a set of blades, and a generator are the components of a wind turbine system that transform the energy of the wind into electric power. The amount of usable wind energy depends on both windspeed as well as the surface of the revolving blades [16, 17]. No need to mention that MPPT is necessary to maximize the use of wind energy. For various wind conditions, these MPPT algorithms monitor the maximum power. The goal of this strategy is to get the most power possible out of the wind turbine system by measuring the output power in real time and monitoring the rate at which power changes with speed. MPPT can be achieved by modifying the converter's duty cycle or rotor speed. The equation below is known as the tip speed ratio (TSR) [18].

$$\lambda = \frac{R\omega}{Velocity} \quad (2)$$

where ω is the electromechanical frequency of the generator and velocity is the wind speed, gives the definition of tip speed ration which is very significant definition as seen in Fig. 3.

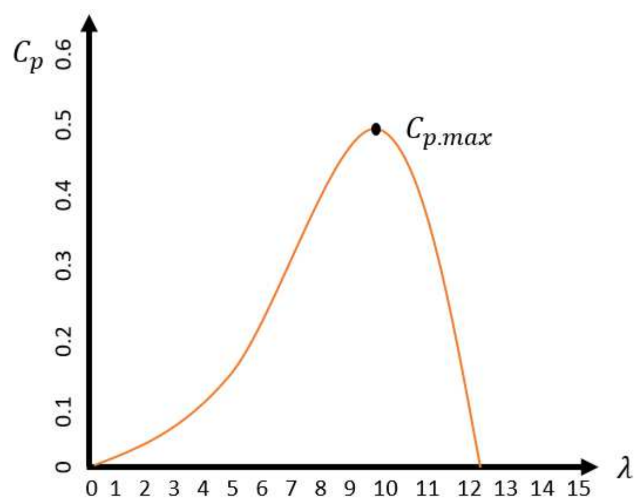

Fig. 3. Cp vs λ

Additionally, the wind turbine's maximum power may be obtained as follow:

$$P_{max} = \frac{1}{2} C_p \rho v^3 \quad (3)$$

where ρ is the air density. According to Fig. 4 and the above equation, the highest amount of power P_{max} is obtained with maximum value of λmax and $Cpmax$ [19].

III. PROPOSED EV CHARGING STATION

The proposed EV charging station in this paper contains two renewable energy sources including wind and solar system. The wind system contains 10 wind turbines and the nameplate capacity for the whole wind system is 600kW. Deimos DT2160 is utilized in this paper with three blades and the rotor diameter is 21 meters, the hub height of 50 meters, with the cut-in wind speed of 3 meter per second and cut-out wind speed of 25 meter per second. Fig. 4, shows the power curve of a turbine.

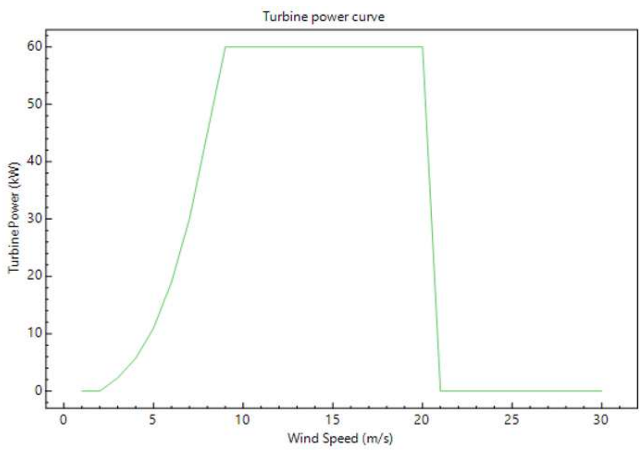

Fig. 4. Turbine power curve

As it can be seen, the output power of each turbine is 60 kW. In addition, the swept area by the blades is 362 square meters. Furthermore, table 1 shows the solar system used in this paper which has the nameplate capacity of 500kW.

TABLE 1. Presented solar system

Cells numbers per module	72	Efficiency	21.08 %
Module Size	1.6*1.14 m	Weight	18.5 kg
Rated power	339.3 W	Tilt	33.5 °
I (SC)	6.54 A	V(OC)	74.6 V
I (MPP)	6.14 A	V (MPP)	62.2 V

According to table 1, each module has 72 cells in series, and the weight of each module is 18.5 kg. the rated power is 339W and the tilt angle for the solar panels is 33.5 degrees which is equal to Lubbock latitude angle. Using solar and wind at the same time increase the reliability of the system. The proposed charging station block diagram is seen in Fig. 5.

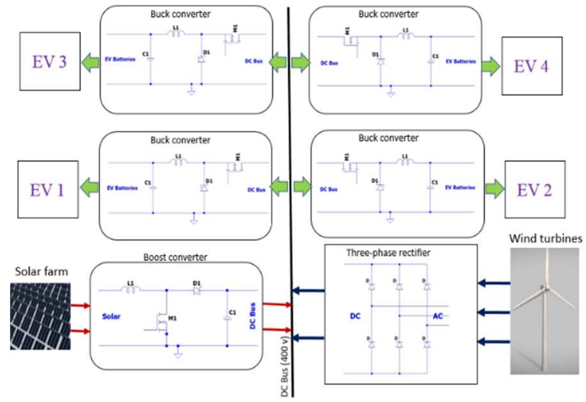

Fig. 5. Charging station block diagram

During the day, solar and wind energy are used to recharge the battery. By stepping up the voltage to 400V using a boost converter, the solar panel's output is managed. Since wind is present at night, it partially charges the DC bus. Then the EVs are charged by battery via utilizing a buck converter. Solar energy is widely available and cost-free, but power stability is challenging. However, it may be effectively made up for by fusing with another energy source. As a result, in this study, solar and wind energy are combined. When there is wind, the battery is charged. When it is hot, a PV module may be used to charge the battery. This research proposes a hybrid system that is unusual in that it generates power from two separate renewable sources. When the hybrid solar/wind system is unable to absorb enough energy, the extra energy stored in the battery is utilized to charge EVs [20, 21].

IV. SIMULATION RESULTS

The proposed system is simulated via LTSPICE, MATLAB, and SAM software. The two significant factors in order to utilize the solar/wind hybrid system is the weather temperature and wind speed. According to the data from Lubbock, the number of sunny days is high and also the wind speed is quite suitable for wind turbines in this area. Fig. 6 depicts the ambient temperature and wind speed of Lubbock.

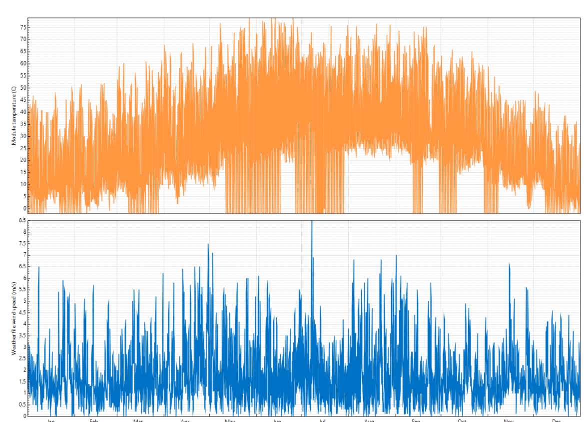

Fig. 6. Ambient temperature and wind speed

Based on the above figure, the highest temperature during the year is 75 degrees Fahrenheit and the average wind speed is around 6 meter per second which is desirable for the turbines. Fig. 7 indicates the monthly extracted power by the proposed wind system.

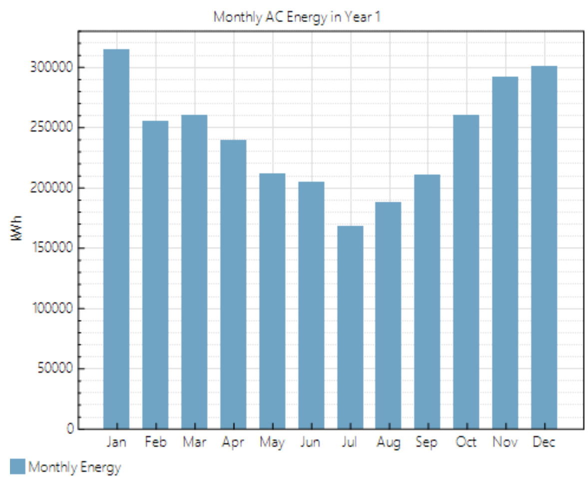

Fig. 7. Monthly extracted power by the proposed wind system

According to Fig. 7, the output wind power is the highest during wintertime and lowest during summertime. For example, in January the output power is around 300000 kWh which is about two times than July for the same year. Also, Fig. 8 demonstrate the monthly extracted power of the proposed solar system in this paper.

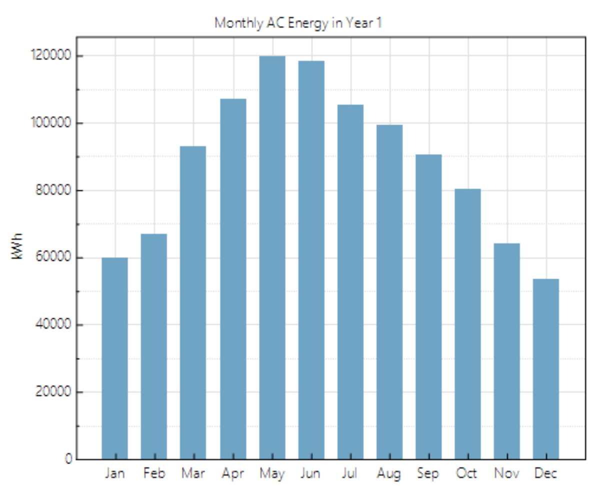

Fig. 8. Monthly extracted power by the proposed solar system

As it can be seen from Fig. 7, the output solar power is the highest during summertime and lowest during wintertime because of two very basic reasons which are the angle of the sun and the number of cloudy days in the winter season. For example, in January the output power is around 60000 kWh which is about the half of the power in July for the same year.

After producing the power from solar system, a boost converter is required to step up the voltage to 400V for the DC bus. Fig. 9 shows the voltage of the solar system and the output voltage of the utilized boost converter.

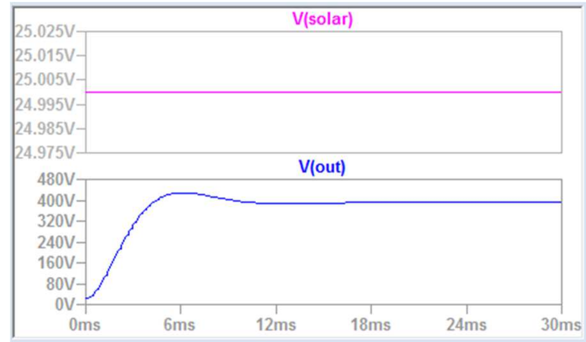

Fig. 9. Output voltage of the utilized boost converter.

Based on Fig. 9, the employed boost converter step up the voltage from 25V to 400V in the proposed DC fast charging station. No need to state that in order to charge the EV batteries a buck converter is required to step down the voltage for the batteries. Fig. 10 shows the output voltage of the buck converter which should be equal to the needed voltage of the EV batteries.

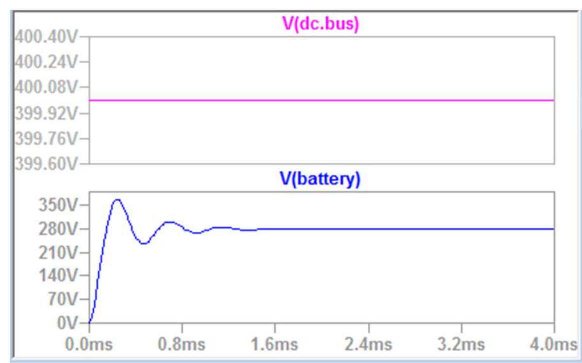

Fig. 10. Output voltage of the utilized buck converter.

According to the figure above, by using a buck converter, the voltage of the DC bus will be changed to the desirable voltage for the EV bank of batteries. Finally, the extracted power for each of the renewable system as well as the whole system is illustrated in Table 2.

TABLE 2. Comparison table for the proposed system

	Wind system	Solar system	Hybrid system
Nameplate capacity	600kW	500kW	1.1MW
Extracted power	2,904,420 kWh	1,057,599 kWh	3,962,019 kWh

Table 2 shows that the total system's extracted power is around 4 GW annually. If each EV takes 30 kWh for charging, then 133000 cars may be charged annually, which is a very huge number.

V. Conclusion

The rise in EV utilization creates a demand for energy that the infrastructure of the present power system cannot provide. In this work, a hybrid renewable energy charging station that requires no external electrical source and has no maintenance costs is presented. The storage batteries are charged by the use of renewable energy sources, and from them, the cars are charged. In these charging stations, the cost of power is lower, and simple to use power is obtained. According to the simulation result, by using just a 600kW wind system consisting of 10 turbines, and the 500kW solar system, the yearly 4 GW power can be achieved in Lubbock which can charge around 133000 EVs which is very remarkable. Furthermore, a very huge amount of CO_2 and SO_2 will be removed from the air.

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